

2022

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: **CACS252** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. If the true value of π is 3.1415926 and its approximate value is given by 3.1428571. Find the absolute and relative errors.
3. What are the advantages of Lagrange's formula over Newton's formula? When Newton's forward interpolation formula is used?
4. Find the polynomial that interpolates $f(0) = 0$, $f(1) = 1$, $f(2) = 0$, $f(3) = 1$, $f(4) = 0$ using Newton divided differences. Use the Newton table to generate the necessary divided differences.
5. Using the trapezium rule, evaluate the integral $\int_1^2 \frac{dx}{x^2 + 6x + 10}$ with 2 and 4 sub intervals. Compare with the exact solution. Comment on the magnitudes of the errors obtained.
6. For solving a linear system, compare Gauss elimination method and Gauss Jordan method. Why Gauss Seidel method is better than Jacobi's iterative method?
7. Using Euler's method find $y(0.2)$ from $\frac{dy}{dx} = x + y$, $y(0) = 1$ with $h = 0.1$.
8. Solve the Poisson equation $\Delta^2 f = 2x^2 y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.

Group C

Attempt any TWO questions [2x10=20]

9. When would we not use N-R method? Find the root of the equation $x^3 - 4x + 1 = 0$, lying in $(0, 1)$ using Bisection method performing 10 iterations.
10. Find the dominant eigenvalue of $A = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$ by power method. Apply Gauss Seidel method to solve system of equations: $6x_1 - 2x_2 + x_3 = 11$; $-2x_1 + 7x_2 + 2x_3 = 5$; $x_1 + 2x_2 - 5x_3 = -1$, with the initial vector of $(0, 0, 0)$.
11. Prepare the multi-step methods available for solving ordinary differential equations. Evaluate the value of y at $x = 0.1$ and 0.2 to 4 decimal places given $\frac{dy}{dx} = x^2 y - 1$ and $y(0) = 1$, using Taylor series method.

